Roll No. | | | | | | | | **Paper Code: MSE 102**

(*M.Tech*) **Odd Mid Semester Examination 2019**

I Semester

Paper Name FEM in Structural Engineering

Time: 1:30 Hours MM: 50

Note:

(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question.
(iii) Total marks for each main question are ten.

Q1.) (2X5=10 Marks)

a. Find the displacement in the two-bar assembly loaded as shown below

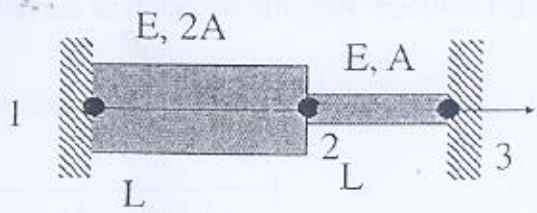


Fig. 1

b. Compute the stresses in the two-bar assembly in fig. 1.

c. Compute the stiffness matrix for the truss below fig.2

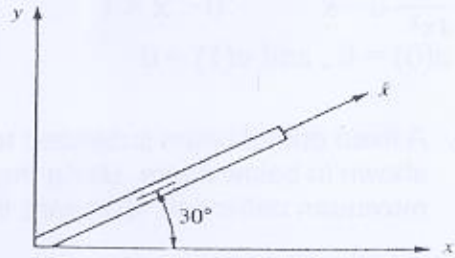
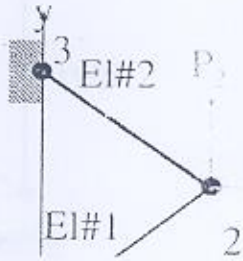


Fig. 2

Q2. (2X5=10 Marks)

a. The length of bars 12 and 23 are equal (L) E: Young's modulus A: Cross sectional area of each bar Solve for (1) d2x and d2y



b. Calculate stresses in each bar of fig. 3 assuming the same data as given in (a) part.

c. Justify how galerkin method is the most accurate method among all the variational method.

(2X5=10 Marks)

Q3.

a Explain the different weighted residual methods. Which one of them is the accurate method for solving differential equation.

b. Calculate the displacements at node 2 and node 3 of the spring given below.

c.

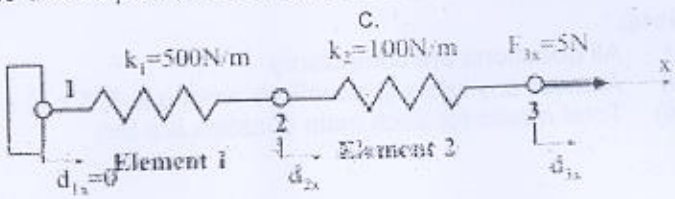


Fig. 4

c . Using any FEM analysis calculate the deflection at point P and compare the solution with exact solution. Assume equation of curve is y=ax(L-x)

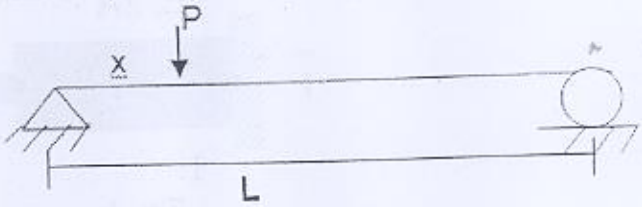


Fig. 5

(2X5=10 Marks)

Q4.

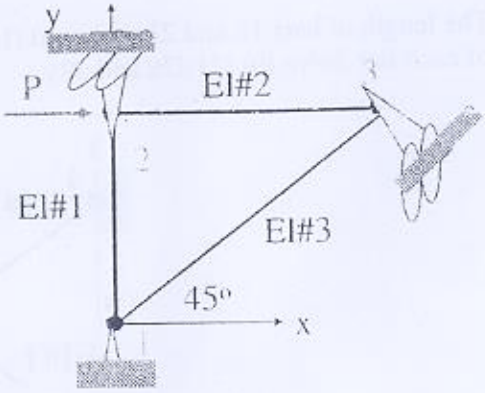
a. Consider the following sample problem Solve by any weighted residual method

$$\frac{-d^2u}{dx^2}-u=-x \qquad 0<x<1$$

$u(0)=0$, and $u(1)=0$

b. A fixed ended beam subjected to uniformly distributed load of intensity p is shown in below figure. Using the Rayleigh-Ritz procedure, compute the maximum deflection. Compare the results with analytical solution.

c. For the plane truss P=1000 kN, L=length of elements 1 and 2 = 1m E=210 GPa , A = $6\times10^{-4}m^2$ for elements 1 and 2, A=2×10^{-4} m^2 for element 3. Determine the unknown displacements.



Q5 (2X5=10 Marks)

a. What are the merits and de merits of finite element method.

b. For the bar shown in figure determine the nodal displacements at node 2 and node 3.

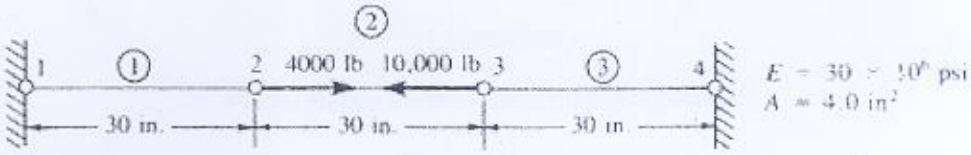


Fig. 7

c. Calculate the stresses and reactions shown in fig.7.

M.Tech

Roll No. | | | | | | | |

Paper Code: MSE-104

M.Tech(Structures)
Odd mid Semester Examination 2019

1st *Semester*
Computer Aided Analysis of Structures

Time:1:30 Hours MM: 50

Note:
(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question.
(iii) Total marks for each main question are ten.

Q1.) (2X5=10 Marks)

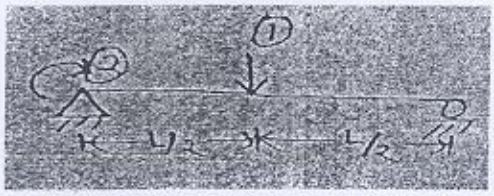
a) Explain the process of idealization of structures?
b) What are the different types of beam elements and boundary conditions in structural elements??
c) Explain the types of loads on the basis of temporal distribution?

Q2. (2X5=10 Marks)

a) Draw a flowchart to explain the steps involved in gauss elimination method?
b) Find all the three matrix norms for given matrix $A=\begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 1 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 1 \end{bmatrix}$?
c) Explain the conditioning of matrices using suitable example?

Q3. (2X5=10 Marks)

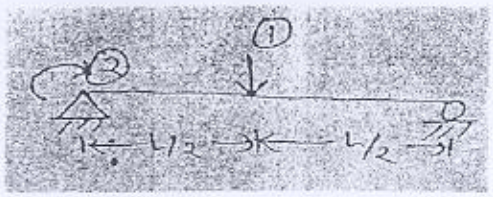
a) Obtain flexibility matrix for following structure with respect to given co-ordinates.



b) Find the deflection at mid span of simply supported beam due to point load at centre using flexibility method?
c) Find the rotation at ends of simply supported beam due to point load at centre using flexibility method?

Q4. (2X5=10 Marks)

a) Obtain stiffness matrix for following structure with respect to given co-ordinates.



b) Find the deflection at mid span of simply supported beam due to point load at centre using stiffness method?

c) Find the rotation at ends of simply supported beam due to point load at centre using stiffness method?

Q5 (2X5=10 Marks)

a) Write all the steps involved in mathematical modelling of structures?

b) What do you understand by static condensation for reduction of dynamic matrices?

c) Discuss any two methods for storage of large sized matrices?

Roll No. | | | | | | | |

Paper Code: MSE 103

(*M.Tech.*)

Mid Semester Examination 2019-20

ISemester

Structural Dynamics

Time: 1:30 Hours

MM: 50

Note: (i) All questions are compulsory.

(ii) Answer any twosub-questionsamong a, b& c in each main question.

(iii)Total marks for each main question are Ten.

Q1. (2 X 5 = 10 Marks)

a) Find out the solution to the following non-homogeneous differential equation.

$$\ddot{u} + 2\xi\omega\dot{u} + \omega_n^2 u = 0$$

b) Find the natural period of the pendulum shown in the figure below for small oscillations about a vertical line passing through the hinge O.

c)Explain the following terms:

i. Subcritical Damping

ii. Critical damping

iii. Overcritical damping

Q2. (2 X 5 = 10 Marks)

a) A block of mass m = 20 Kg is resting on a flat surface and is restrained by a spring of stiffness k = 2500N/m. the coefficient of friction is μ = 0.2. The mass is displaced laterally by 20mm and released instantaneously in the positive direction. The maximum displacement after 10 cycles is 2 mm in the positive direction. Determine the time required for this to occur.

b) Consider a diver weighing 100kg at the end of a diving board that cantilevers out 4ft. The diver oscillates at a frequency of 3Hz. What is the flexural rigidity EI of the diving board?

c) What is animpulse load? Explain with the help of graphs.

Q3. (2 X 5 = 10 Marks)

a) What do you understand by damping? What the various types of damping?

b) Explain D'Alembert's principle. Also explain Dynamic equilibrium.

c) Explain the following terms:

i. Simple Harmonic motion

ii. Logarithmic Decrement

iii. Overdamped systems

iv. Natural frequency of vibration

v. Free vibration

Q4. (2 X 5 = 10 Marks)

a) What are the properties of [M], [K]&[C] matrices?

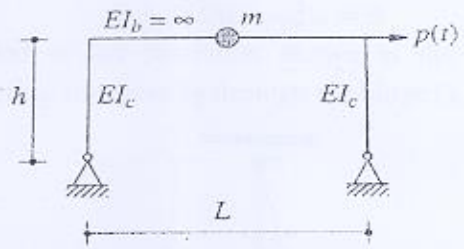
b) Explain dynamic amplification factors R_d, R_v and R_a.

c) Find out the solution to the following linear, homogeneous differential equation with constant coefficients:

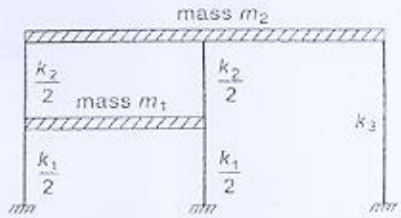
$$m\ddot{u} + ku = p_0 cos\omega t$$

Q5. (2 X 5 = 10 Marks)

a) Derive the equation of motion for the frame shown in figure below. c=0.



b) For the system shown below formulate the equation of motion governing the undamped free vibration.



c) How is the frequency equation derived for a multi degree of freedom system?